Numerical Analysis Assignment 4

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1

Let

$$A = \begin{bmatrix} 4.00 & 2.00 & 0.00 \\ 4.00 & 4.00 & 2.00 \\ 2.00 & 2.00 & 3.00 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

Since

$$L = \begin{bmatrix} 1.00 & 0.00 & 0.00 \\ 1.00 & 1.00 & 0.00 \\ 0.50 & 0.50 & 1.00 \end{bmatrix} \text{ and } U = \begin{bmatrix} 4.00 & 2.00 & 0.00 \\ 0.00 & 2.00 & 2.00 \\ 0.00 & 0.00 & 2.00 \end{bmatrix}$$

such that

$$A = LU$$

Then,

$$A\vec{x} = \vec{b} \Longleftrightarrow \begin{cases} L\vec{c} = \vec{b}, \\ U\vec{x} = \vec{c} \end{cases}$$

The first step of back substitution is

$$\begin{aligned} c_1 =& l_{11}b_1 = 2, \\ c_2 =& (b_1 - l_{21}c_1)/l_{21} = 2, \\ c_3 =& (b_3 - l_{31}c_1 - l_{32}c_2)/l_{33} = 4. \end{aligned}$$

The second step of back substitution is

$$\begin{aligned} x_3 =& c_3/u_{33} = 1, \\ x_2 =& (b_2 - u_{23}x_3)/u_{22} = -1, \\ x_1 =& (b_1 - u_{13}x_3 - u_{12}x_2)/u_{11} = 2. \end{aligned}$$

2

(a)

Apply Gaussian elimination to the argument matrix,

$$\begin{bmatrix} 0.001 & 2.000 & 3.000 & 1.000 \\ -1.000 & 3.712 & 4.623 & 2.000 \\ -2.000 & 1.072 & 5.643 & 3.000 \end{bmatrix} \xrightarrow{r_2+1000r_1 \to r_2} \begin{bmatrix} 0.001 & 2.000 & 3.000 & 1.000 \\ 0.000 & 2003.712 & 3004.623 & 1002.000 \\ -2.000 & 1.072 & 5.643 & 3.000 \end{bmatrix} \xrightarrow{r_3+2000r_1 \to r_3}$$

$$\begin{bmatrix} 0.001 & 2.000 & 3.000 & 1.000 \\ 0.000 & 2003.712 & 3004.623 & 1002.000 \\ 0.000 & 4001.072 & 6005.643 & 2003.000 \end{bmatrix} \xrightarrow{r_3-1.9968r_2 \to r_3} \begin{bmatrix} 0.001 & 2.000 & 3.000 & 1.000 \\ 0.000 & 2003.712 & 3004.623 & 1002.000 \\ 0.000 & 0.000 & 6.012 & 2.206 \end{bmatrix}$$

Then back substitution,

$$x_3 = b_3/a_{33} = 0.3669,$$
$$x_2 = (b_2 - a_{23}x_3)/a_{22} = -0.0501,$$
$$x_1 = (b_1 - a_{12}x_2 - a_{13}x_3)/a_{11} = -0.5000.$$

Apply Gaussian elimination with partial pivoting to the argument matrix,

$$\begin{bmatrix} 0.001 & 2.000 & 3.000 & 1.000 \\ -1.000 & 3.712 & 4.623 & 2.000 \\ -2.000 & 1.072 & 5.643 & 3.000 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ -1.000 & 3.712 & 4.623 & 2.000 \\ 0.001 & 2.000 & 3.000 & 1.000 \end{bmatrix} \xrightarrow{r_2 - 0.5r_1 \to r_2}$$
$$\begin{bmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ 0.000 & 3.176 & 1.802 & 0.500 \\ 0.001 & 2.000 & 3.000 & 1.000 \end{bmatrix} \xrightarrow{r_3 - 0.005r_1 \to r_3} \begin{bmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ 0.000 & 3.176 & 1.802 & 0.500 \\ 0.000 & 2.001 & 3.003 & 1.002 \end{bmatrix}$$
$$\xrightarrow{r_3 - 0.6299r_2 \to r_3} \begin{bmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ 0.000 & 3.176 & 1.802 & 0.500 \\ 0.000 & 0.000 & 1.868 & 0.687 \end{bmatrix}$$

Then back substitution,

$$\begin{aligned} x_3 =& b_3/a_{33} = 0.3678, \\ x_2 =& (b_2 - a_{23}x_3)/a_{22} = -0.05125, \\ x_1 =& (b_1 - a_{12}x_2 - a_{13}x_3)/a_{11} = -0.4897. \end{aligned}$$

(b)

The error of Gaussian elimination is

$$e_1 = ||\mathbf{x} - \mathbf{x}^*|| = 1.2196.$$

The error of Gaussian elimination with partial pivoting is

$$e_2 = ||\mathbf{x}_p - \mathbf{x}^*|| = 1.2129.$$

3

Suppose that a matrix E is an equivalent matrix representation of the row operations applied to A, such that

$$EA = \begin{bmatrix} a_{11} & \mathbf{a}^T \\ \mathbf{0} & B \end{bmatrix}$$

Then, the matrix E and A can be written as

$$E = \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{x} & I \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & \vec{\alpha}^T \\ \vec{\alpha} & \mathcal{A} \end{bmatrix}$$

then

$$B = \mathbf{x}\vec{\alpha}^T + \mathcal{A}.$$

Since

$$\begin{bmatrix} a_{11} & \mathbf{a}^T \\ \mathbf{0} & B \end{bmatrix}^T = \begin{bmatrix} a_{11} & \mathbf{0}^T \\ \mathbf{a} & B^T \end{bmatrix} = AE^T = \begin{bmatrix} a_{11} & \vec{\alpha}^T \\ \vec{\alpha} & \mathcal{A} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{x}^T \\ \mathbf{0} & I \end{bmatrix}$$

So,

$$B^T = \vec{\alpha}\mathbf{x}^T + \mathcal{A}^T = (\vec{\alpha}\mathbf{x}^T + \mathcal{A}^T)^T = \mathbf{x}\vec{\alpha}^T + \mathcal{A} = B.$$

Therefore, if A is a symmetric matrix, then B is symmetric, too.

4

Assume that the computation duration is proportional to how many operations it needs. Suppose there is a linear function $t = f(n)$ between computation duration t and the number of operations n.

Since the costs of Gaussian elimination and LU decomposition are similar. Assume that the 2000×2000 linear system is solved by Gaussian elimination. The cost of Gaussian elimination is

$$f_{ge}(n) = \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{7}{6}n.$$

The cost of substitution is $g(n) = n^2$. Since there is only back-substitution for Gaussian elimination]Then, the cost of Gaussian elimination is

$$f_{ge}(2000) + g(2000) = 5339331000.$$

respectively. Then the computer can complete $5339331000/0.1 = 53393310000$ operations. The computational cost of 100 8000×8000 linear systems by LU decomposition is

$$\frac{2}{3}8000^3 - \frac{1}{2}8000^2 - \frac{1}{6}8000 + 101 \times 8000^2 = 347765332000.$$

The computational duration is $347765332000/53393310000 = 6.5133$ seconds.

5

Since A is strictly column diagonally dominant, so a_{11} is the "largest"[1] element. So, no row interchange takes place in the first step of Gaussian elimination.

After the first step of Gaussian elimination, for $2 \leq i, j \leq n$,

$$a'_{ij} = a_{ij} - \frac{a_{i,1}}{a_{11}}a_{1,j},$$

where a'_{ij} is the corresponding element of the matrix A' that is generated by applying the first step of Gaussian elimination to A. By triangle inequality, for $2 \leq i, j \leq n$,

$$|a_{i1}| - \left|\frac{a_{i,1}}{a_{11}}a_{1,j}\right| \leq |a'_{i,j}| \leq |a_{i1}| + \left|\frac{a_{i,1}}{a_{11}}a_{1,j}\right|.$$

For nth column of transferred matrix A', there is

$$\begin{aligned} |a_{n,n}| - |a_{k,n}| &> \sum_{i=1, i\neq n}^{m} |a_{i,n}| - |a_{k,n}| \\ &> |a_{1,n}| \\ &> \frac{|a_{n,1}| + |a_{k,1}|}{|a_{11}|}|a_{1,n}| \end{aligned}$$

for $k \neq 1$ and n. So,

$$|a_{n,n}| - \frac{|a_{n,1}|}{|a_{11}|}|a_{1,n}| > |a_{k,n}| + \frac{|a_{n,1}|}{|a_{11}|}|a_{1,n}|$$

Therefore, by the triangle inequality,

$$|a'_{n,n}| \geq |a_{n,n}| - \frac{|a_{n,1}|}{|a_{11}|}|a_{1,n}| > |a_{k,n}| + \frac{|a_{n,1}|}{|a_{11}|}|a_{1,n}| \geq |a'_{k,n}|.$$

It is proved that if the diagonal element is the largest in its column, then after one step of Gaussian elimination, the diagonal element is still the largest in its column. In subsequent steps of Gaussian elimination, no row interchange takes place.

[1] The "largest" means the absolute value is largest